Airbnb Price Analysis

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Introduction

This project analyzes Airbnb listing data to understand the factors that influence pricing in Bangkok, Thailand. Airbnb listings vary widely in price, room type, and customer demand, making it useful to examine which factors are most strongly associated with listing prices.

While it is commonly assumed that higher demand leads to higher prices, this study examines whether customer demand actually influences Airbnb pricing compared to structural listing characteristics.

The research question for this study is:

How do room type and customer demand (number of reviews) influence Airbnb pricing in Bangkok?

Data

The dataset is sourced from Inside Airbnb and includes listing-level data such as price, room type, and number of reviews.

```
airbnb <- read_csv("/Users/mrcolli/Desktop/DSA 8660/listings-B.csv")
```

```
## Rows: 28806 Columns: 79
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (24): listing_url, source, name, description, neighborhood_overview, pi...
## dbl  (42): id, scrape_id, host_id, host_listings_count, host_total_listings_...
## lgl   (8): host_is_superhost, host_has_profile_pic, host_identity_verified, ...
## date  (5): last_scraped, host_since, calendar_last_scraped, first_review, la...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(airbnb)
```

```
## Rows: 28,806
## Columns: 79
## $ id                                          <dbl> 27934, 27979, 28745, 4751~
## $ listing_url                                 <chr> "https://www.airbnb.com/r~
## $ scrape_id                                   <dbl> 2.025093e+13, 2.025093e+1~
```

```
## $ last_scraped                    <date> 2025-09-27, 2025-09-27, ~
## $ source                          <chr> "city scrape", "previous ~
## $ name                            <chr> "Nice room with superb ci~
## $ description                     <chr> "Our cool and comfortable~
## $ neighborhood_overview           <chr> "It is very center of Ban~
## $ picture_url                     <chr> "https://a0.muscache.com/~
## $ host_id                         <dbl> 120437, 120541, 123784, 2~
## $ host_url                        <chr> "https://www.airbnb.com/u~
## $ host_name                       <chr> "Nuttee", "Emy", "Familyr~
## $ host_since                      <date> 2010-05-08, 2010-05-08, ~
## $ host_location                   <chr> "Bangkok, Thailand", "Ban~
## $ host_about                      <chr> "Hi All, I am nuttee patr~
## $ host_response_time              <chr> "N/A", "N/A", "N/A", "N/A~
## $ host_response_rate              <chr> "N/A", "N/A", "N/A", "N/A~
## $ host_acceptance_rate            <chr> "67%", "N/A", "N/A", "N/A~
## $ host_is_superhost               <lgl> FALSE, FALSE, FALSE, FALS~
## $ host_thumbnail_url              <chr> "https://a0.muscache.com/~
## $ host_picture_url                <chr> "https://a0.muscache.com/~
## $ host_neighbourhood              <chr> "Victory Monument", NA, "~
## $ host_listings_count             <dbl> 1, 2, 1, 1, 1, 4, 2, 3, 1~
## $ host_total_listings_count       <dbl> 2, 4, 1, 2, 1, 6, 2, 9, 4~
## $ host_verifications              <chr> "['email', 'phone']", "['~
## $ host_has_profile_pic            <lgl> TRUE, TRUE, TRUE, FALSE, ~
## $ host_identity_verified          <lgl> TRUE, FALSE, FALSE, TRUE,~
## $ neighbourhood                   <chr> "Samsen Nai, Bangkok, Tha~
## $ neighbourhood_cleansed          <chr> "Ratchathewi", "Bang Na",~
## $ neighbourhood_group_cleansed    <lgl> NA, NA, NA, NA, NA, NA, N~
## $ latitude                        <dbl> 13.75983, 13.66818, 13.75~
## $ longitude                       <dbl> 100.5413, 100.6167, 100.6~
## $ property_type                   <chr> "Entire condo", "Private ~
## $ room_type                       <chr> "Entire home/apt", "Priva~
## $ accommodates                    <dbl> 2, 2, 2, 1, 2, 2, 2, 5, 1~
## $ bathrooms                       <dbl> 1.5, NA, NA, NA, 1.0, 1.0~
## $ bathrooms_text                  <chr> "1.5 baths", "1 bath", "1~
## $ bedrooms                        <dbl> 1, NA, 1, 4, 1, 1, NA, 3,~
## $ beds                            <dbl> 1, NA, NA, 1, 1, 1, NA, 3~
## $ amenities                       <chr> "[\"Free parking on premi~
## $ price                           <chr> "$1,595.00", NA, NA, "$4,~
## $ minimum_nights                  <dbl> 15, 1, 60, 3, 14, 2, 1, 2~
## $ maximum_nights                  <dbl> 240, 730, 730, 730, 365, ~
## $ minimum_minimum_nights          <dbl> 30, 1, 60, 3, 14, 2, 1, 2~
## $ maximum_minimum_nights          <dbl> 30, 1, 60, 3, 14, 2, 1, 2~
## $ minimum_maximum_nights          <dbl> 240, 730, 730, 730, 365, ~
## $ maximum_maximum_nights          <dbl> 240, 730, 730, 730, 365, ~
## $ minimum_nights_avg_ntm          <dbl> 30, 1, 60, 3, 14, 2, 1, 2~
## $ maximum_nights_avg_ntm          <dbl> 240, 730, 730, 730, 365, ~
## $ calendar_updated                <lgl> NA, NA, NA, NA, NA, NA, N~
## $ has_availability                <lgl> TRUE, NA, NA, NA, TRUE, T~
## $ availability_30                 <dbl> 27, 0, 0, 30, 30, 30, 0, ~
## $ availability_60                 <dbl> 57, 0, 0, 60, 60, 60, 0, ~
## $ availability_90                 <dbl> 87, 0, 0, 90, 90, 90, 0, ~
## $ availability_365                <dbl> 362, 0, 0, 365, 365, 365,~
## $ calendar_last_scraped           <date> 2025-09-27, 2025-09-27, ~
## $ number_of_reviews               <dbl> 65, 0, 0, 0, 1, 39, 0, 14~
```

```
## $ number_of_reviews_ltm                          <dbl> 0, 0, 0, 0, 0, 5, 0, 0, 0~
## $ number_of_reviews_l30d                         <dbl> 0, 0, 0, 0, 0, 1, 0, 0, 0~
## $ availability_eoy                               <dbl> 93, 0, 0, 96, 96, 96, 0, ~
## $ number_of_reviews_ly                           <dbl> 1, 0, 0, 0, 0, 5, 0, 0, 0~
## $ estimated_occupancy_l365d                      <dbl> 0, 0, 0, 0, 0, 30, 0, 0, ~
## $ estimated_revenue_l365d                        <dbl> 0, NA, NA, 0, 0, 41040, N~
## $ first_review                                   <date> 2012-04-07, NA, NA, NA, ~
## $ last_review                                    <date> 2024-09-17, NA, NA, NA, ~
## $ review_scores_rating                           <dbl> 4.86, NA, NA, NA, 5.00, 4~
## $ review_scores_accuracy                         <dbl> 4.95, NA, NA, NA, 5.00, 4~
## $ review_scores_cleanliness                      <dbl> 4.82, NA, NA, NA, 5.00, 4~
## $ review_scores_checkin                          <dbl> 4.97, NA, NA, NA, 5.00, 4~
## $ review_scores_communication                    <dbl> 4.91, NA, NA, NA, 5.00, 4~
## $ review_scores_location                         <dbl> 4.66, NA, NA, NA, 5.00, 4~
## $ review_scores_value                            <dbl> 4.75, NA, NA, NA, 5.00, 4~
## $ license                                        <lgl> NA, NA, NA, NA, NA, NA, N~
## $ instant_bookable                               <lgl> FALSE, FALSE, FALSE, FALS~
## $ calculated_host_listings_count                 <dbl> 1, 2, 1, 1, 1, 4, 1, 1, 1~
## $ calculated_host_listings_count_entire_homes    <dbl> 1, 1, 0, 1, 0, 0, 0, 1, 0~
## $ calculated_host_listings_count_private_rooms   <dbl> 0, 1, 1, 0, 1, 2, 1, 0, 1~
## $ calculated_host_listings_count_shared_rooms    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ reviews_per_month                              <dbl> 0.40, NA, NA, NA, 0.01, 0~
```

Method

The data was cleaned by converting the price variable from character format to numeric format and removing missing or invalid price values. Summary statistics were used to understand the price distribution. A log-scaled histogram was used to display the full distribution of prices, while a filtered histogram was used to better visualize the central range of prices. A boxplot and scatterplot were also used to examine the relationships between price, room type, and number of reviews.

```
airbnb_clean <- airbnb %>%
  mutate(price = as.numeric(gsub("[$,]", "", price))) %>%
  filter(!is.na(price), price > 0)

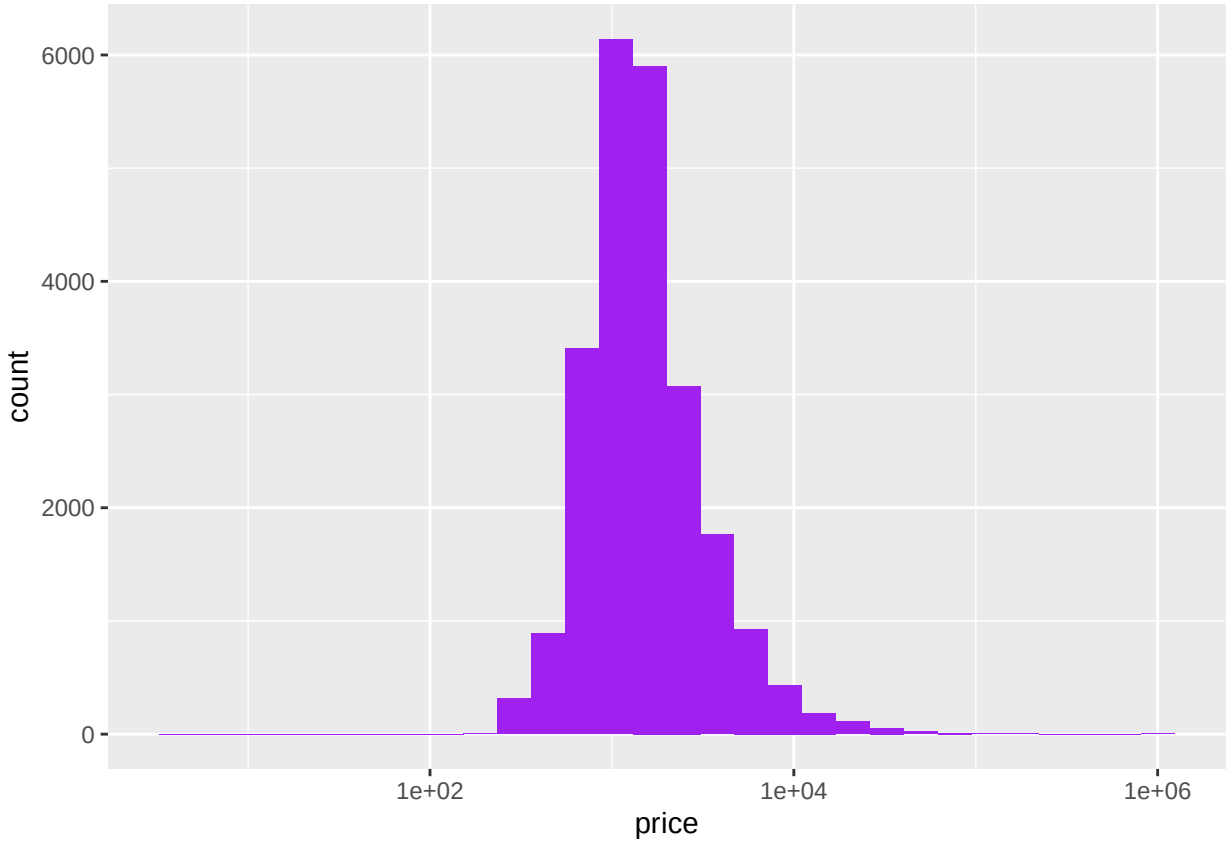
summary(airbnb_clean$price)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       4     923    1379    2529    2207 1000000
```

Results

Price Distribution (Log Scale)

```
ggplot(airbnb_clean, aes(price)) +
  geom_histogram(fill = "purple", bins = 30) +
  scale_x_log10()
```

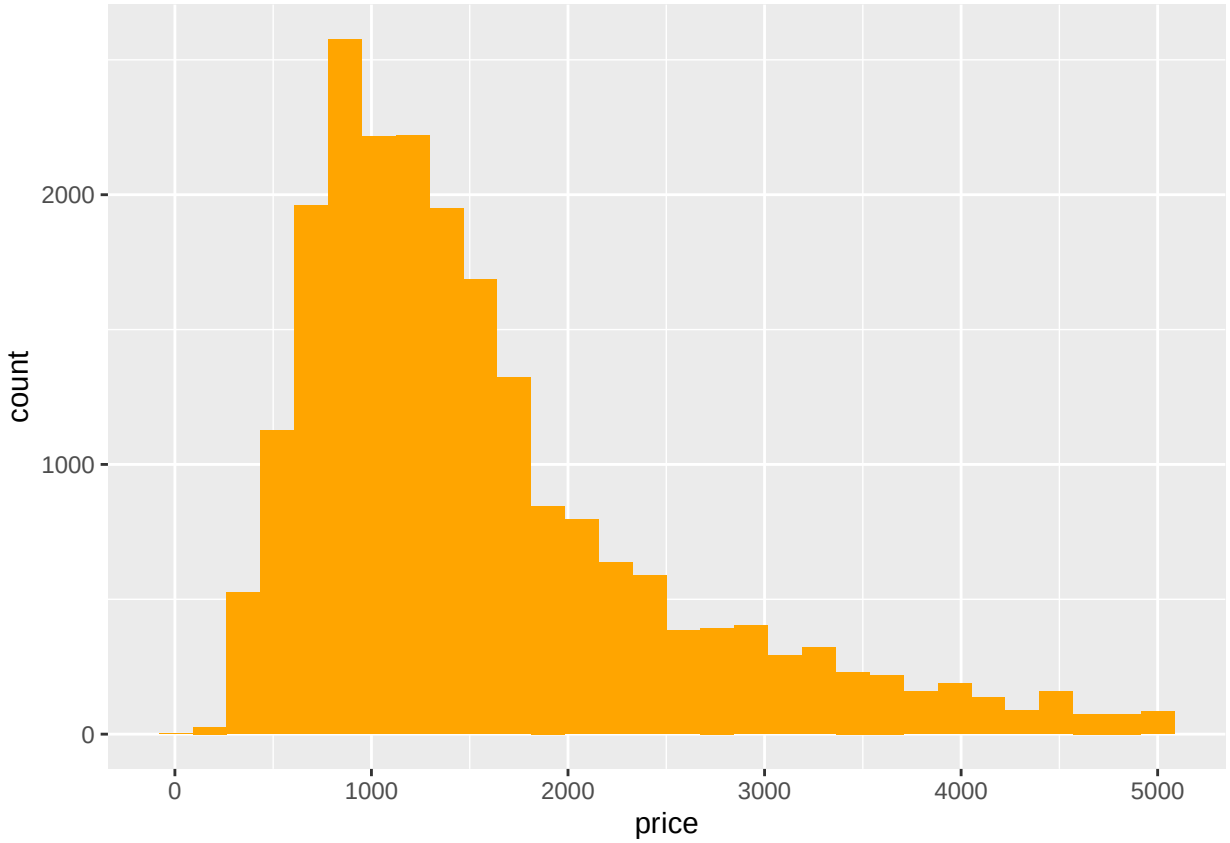


The log-scaled histogram shows that Airbnb prices in Bangkok are strongly right-skewed. Most listings fall within a moderate price range, while a small number of listings are priced much higher.

Price Distribution (Filtered)

```
airbnb_clean2 <- airbnb_clean %>%
  filter(price <= 5000)

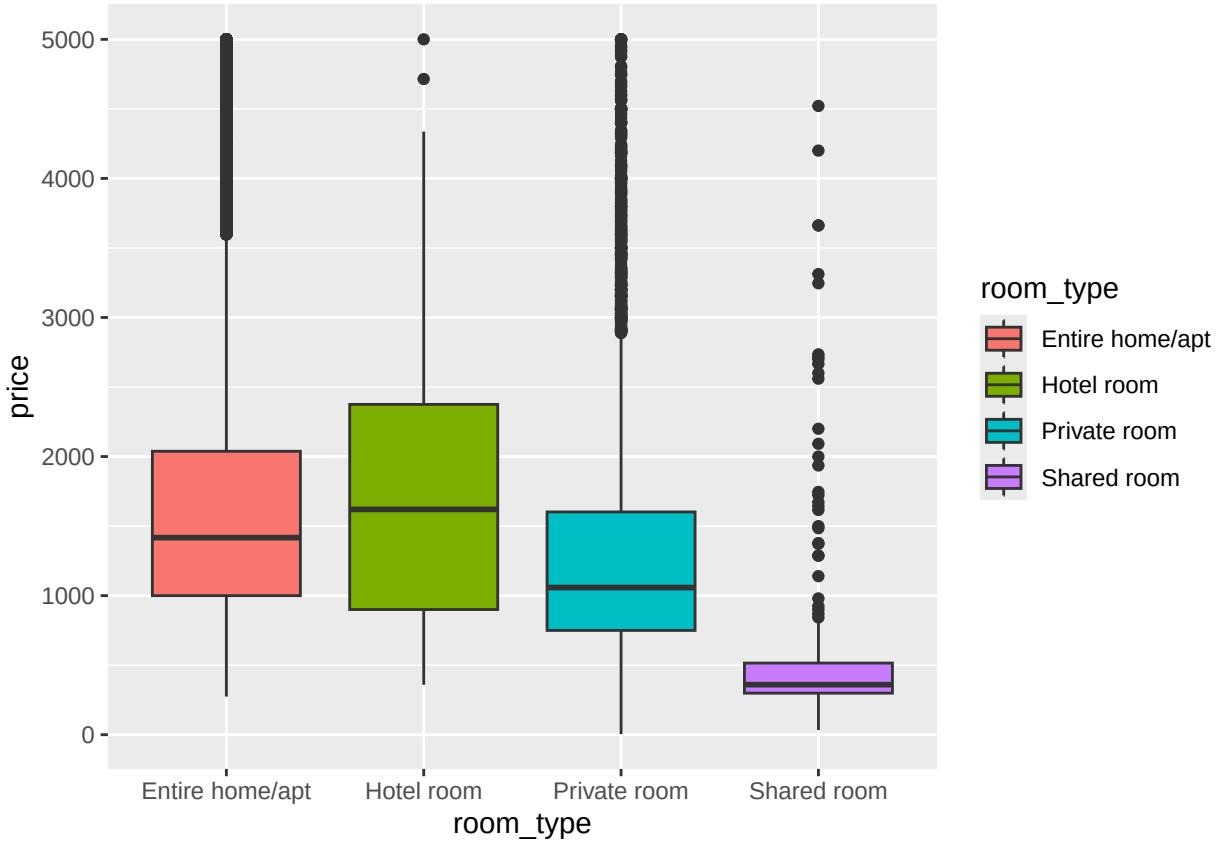
ggplot(airbnb_clean2, aes(x = price)) +
  geom_histogram(fill = "orange", bins = 30)
```



After filtering prices above 5,000, the distribution becomes easier to interpret. Most listings are concentrated in the lower to mid-price range.

Price by Room Type

```
ggplot(airbnb_clean2, aes(x = room_type, y = price, fill = room_type)) +
  geom_boxplot()
```



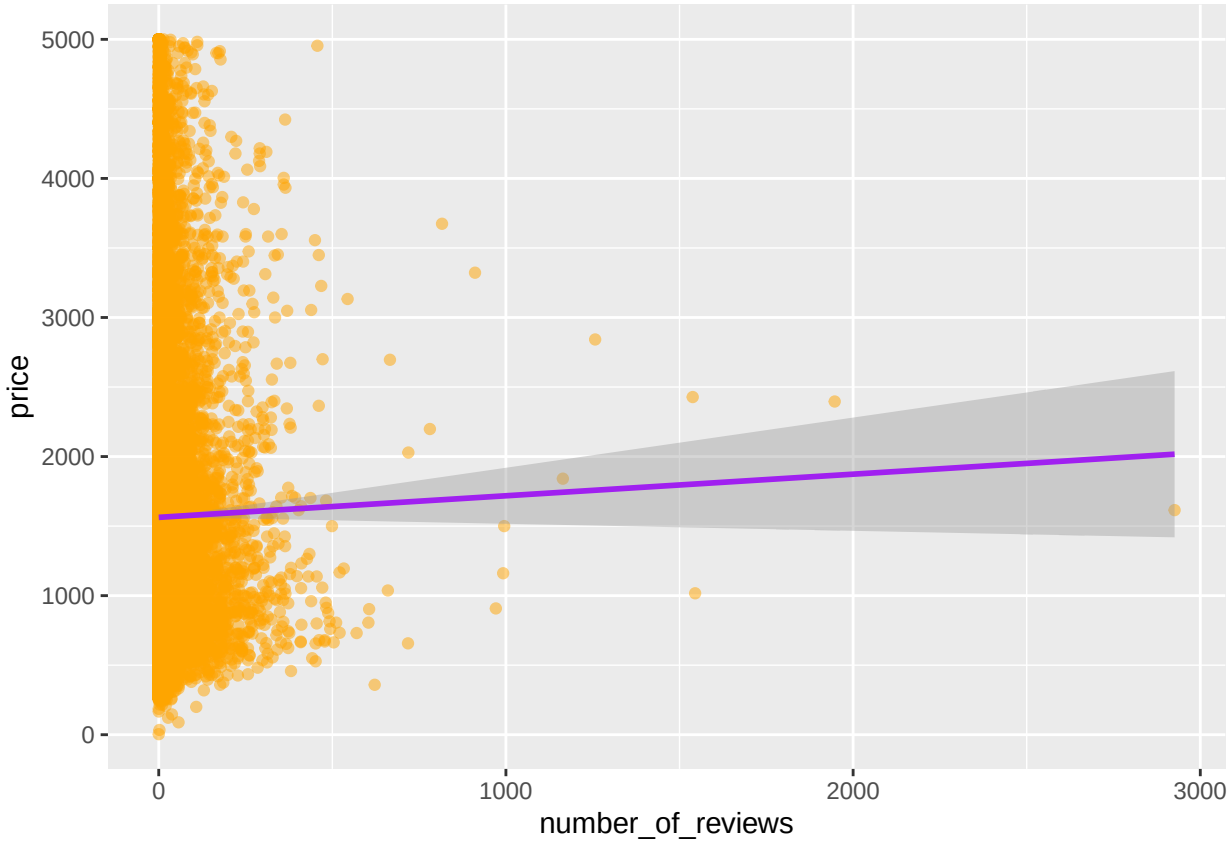
The boxplot shows clear differences in price by room type. Hotel rooms and entire homes tend to have higher prices, while private rooms and shared rooms are generally less expensive. This pricing difference is likely driven by the level of privacy and amenities offered. Entire homes and hotel rooms provide full access to the property and additional services, which increases their value. In contrast, private and shared rooms offer fewer amenities and less privacy, resulting in lower prices.

Price vs. Reviews

This finding challenges the common assumption that higher demand leads to higher prices.

```
ggplot(airbnb_clean2, aes(x = number_of_reviews, y = price)) +
  geom_point(color = "orange", alpha = 0.5) +
  geom_smooth(method = "lm", color = "purple")
```

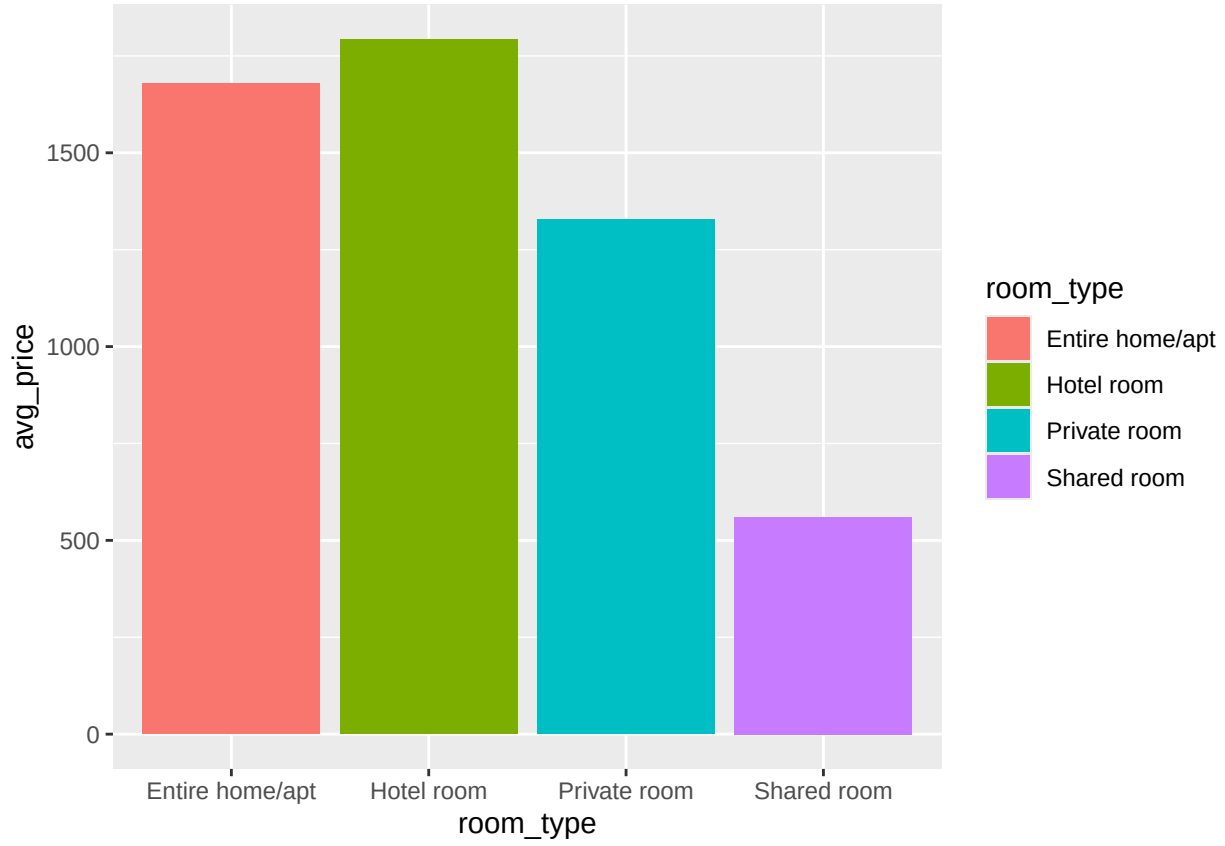
```
## `geom_smooth()` using formula = 'y ~ x'
```



The scatterplot shows a weak positive relationship between number of reviews and price. Listings with more reviews tend to have slightly higher prices, but the relationship is not strong. This suggests that customer demand alone does not determine Airbnb pricing. This pricing difference is likely driven by the level of privacy and amenities offered. Entire homes and hotel rooms provide full access to the property and additional services, which increases their value. In contrast, private and shared rooms offer fewer amenities and less privacy, resulting in lower prices.

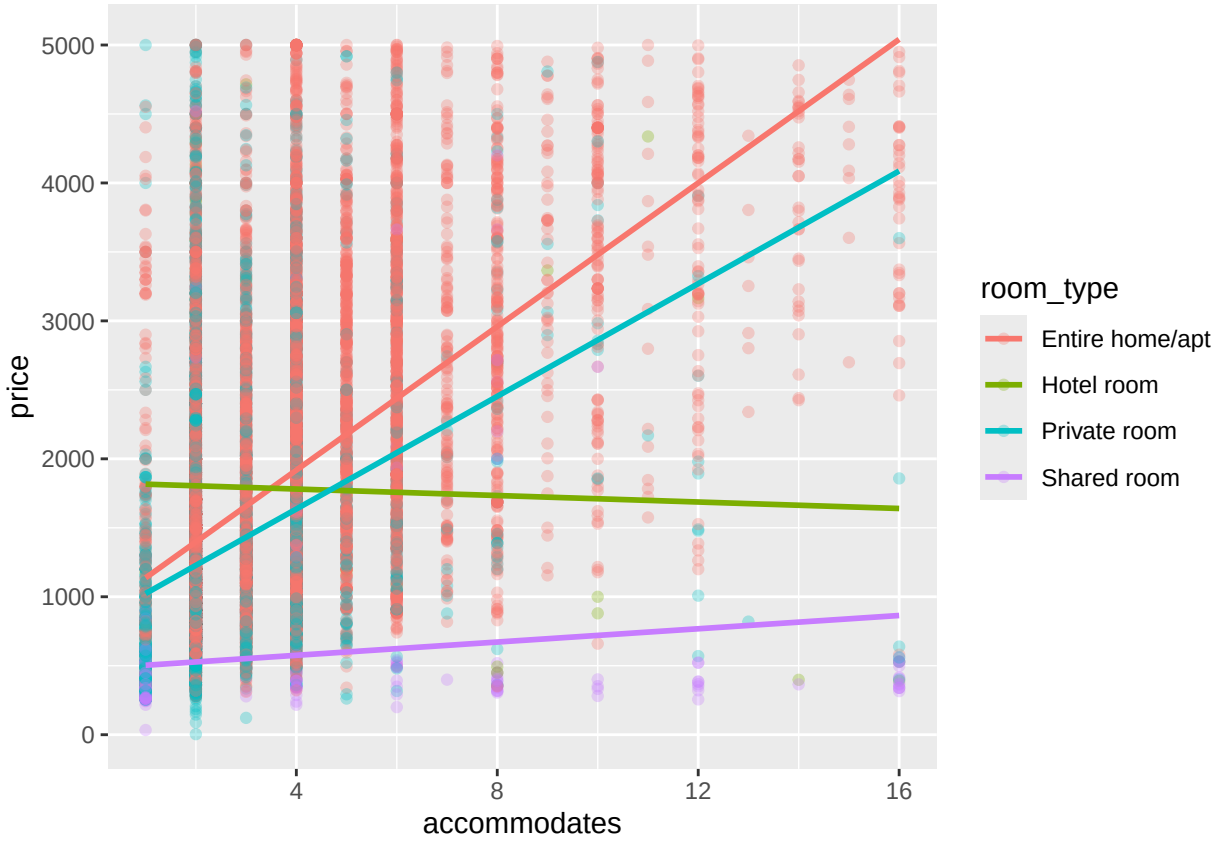
Although listings with more reviews indicate higher demand, price does not increase proportionally. This suggests that pricing is determined more by structural features such as room type and amenities rather than customer demand alone.

```
airbnb_clean2 %>%
  group_by(room_type) %>%
  summarise(avg_price = mean(price)) %>%
  ggplot(aes(room_type, avg_price, fill = room_type)) +
  geom_col()
```




```
ggplot(airbnb_clean2, aes(accommodates, price, color = room_type)) +
  geom_point(alpha = .3) +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Conclusion

This analysis reveals that Airbnb pricing in Bangkok is primarily driven by structural listing characteristics rather than customer demand. While listings with more reviews indicate higher popularity, they do not consistently command higher prices. This challenges the common assumption that demand alone determines Airbnb pricing.

Room type plays a dominant role in determining price, with entire homes and hotel-style accommodations generally priced higher due to increased privacy, amenities, and perceived value. In contrast, private and shared rooms tend to be priced lower and may rely more on competitive pricing strategies.

From a managerial perspective, Airbnb hosts should prioritize listing characteristics such as property type, space, and amenities when setting prices. Since review count shows only a weak relationship with price, hosts should not rely solely on demand metrics when determining pricing strategies. Instead, pricing decisions should be based on the overall value and features offered by the listing.